


**Department of Information Technology**
**Faculty of Computing**
**International Campus of Science and Technology**

**Higher Diploma in Information Technology**
**Batch – 01, 2026**

# SMART HOSPITAL SYSTEM

*An Integrated IoT, Virtual Reality and Augmented Reality Simulation for Patient-Ward Monitoring and Care*

| | |
|---|---|
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## Declaration

I declare that this project report is my own work, prepared for the Emerging Technologies (HDIT21132) Integrated Simulation Project. The Cisco Packet Tracer network, the Delightex EDU virtual environment, the MyWebAR augmented-reality scene, and all screenshots, descriptions, and evaluations presented in this report were designed, built, and tested by me. All external tools and sources referenced are acknowledged in the References section.

Student:

Signature: _____

Date: _____

## Acknowledgement

I would like to thank my lecturer for the guidance and feedback provided throughout the Emerging Technologies module, which helped shape the design of the IoT network, the virtual ward walkthrough, and the augmented-reality scene described in this report. I also thank the International Campus of Science and Technology (ICST) for providing access to the software platforms used in this project. Finally, I appreciate the encouragement of my family and classmates during the design, simulation, and documentation stages of this work.

## Abstract

Patients in a hospital ward are frequently unable to respond to hazards on their own, so a delay of even a few minutes in noticing smoke, an overheated room, or an unattended emergency can be far more serious than the same delay would be in an ordinary home. This project designs and simulates an integrated Smart Hospital System that combines the Internet of Things (IoT), Virtual Reality (VR), and Augmented Reality (AR) to address that gap.

The IoT layer was built in Cisco Packet Tracer around a Home Gateway that wirelessly connects a monitoring laptop, a smoke detector, a combined temperature-and-humidity ("Humiture") monitor, a hospital emergency siren, a smart light, and a smart oscillating fan. Eight enabled automation conditions were configured so that a single smoke event simultaneously raises the siren, switches on the light, and drives the fan to help clear the room, while a separate humidity/temperature condition manages patient comfort independently. The VR layer was created in Delightex EDU as a fully furnished ward containing a two-bed patient room, an en-suite washroom, a reception desk, and three visitor waiting zones. Twenty-six individual assets — beds, doctors, a nurse, monitors, the siren, fire extinguishers, the air-conditioner, lamps, the reception laptop, and more — were wired with CoBlocks click events so that each one opens a titled information panel, turning the walkthrough into a self-guided orientation tour. The AR layer was created in MyWebAR using three 3D models — a reception desk, a bedside lamp, and a reception laptop — each configured to play an on-start animation and to open the VR ward tour in the browser when tapped, directly linking the mobile AR layer back into the immersive VR layer.

Together, the three layers demonstrate three complementary roles: IoT senses hazards and reacts automatically, VR lets staff or students explore and learn the ward layout without any risk to real

patients, and AR gives a fast, mobile-friendly bridge between a physical object and the same virtual information. Testing confirmed that every minimum requirement of the brief was met and, in the case of the VR interactivity and IoT automation logic, substantially exceeded.

Keywords: Smart Hospital, Internet of Things, Cisco Packet Tracer, Virtual Reality, Delightex EDU, Augmented Reality, MyWebAR, Patient Safety, Automation.

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# Chapter 1 — Introduction

***Problem context, objectives, scope, and report organization***

## 1.1 Background

Emerging technologies are changing how buildings and the people inside them are monitored and supported. The Internet of Things lets sensors and appliances share data and react automatically. Virtual Reality lets a proposed environment be explored safely before it is built or before new staff ever set foot in it. Augmented Reality overlays digital information on a real-world view through a phone or tablet. In a hospital ward these three technologies matter more than almost anywhere else, because the people inside the environment — patients — are often unable to act on their own behalf.

## 1.2 Problem Background

A conventional hospital ward relies heavily on nurses noticing problems in person: walking the corridor, checking on patients, and responding to call bells. A nurse attending to one patient cannot simultaneously watch a smoke detector, a thermostat, and a call button in another room. New nursing students or facilities staff are also normally trained on the live ward itself, which is disruptive to patients and carries real risk if a trainee makes a mistake with actual equipment. A simulation-based approach removes both problems: automation reduces the time between a hazard occurring and a response beginning, and a virtual/augmented environment lets people learn the layout and the equipment without any risk to a real patient.

## 1.3 Problem Statement

There is a need for an integrated smart-ward solution that can (a) continuously monitor environmental conditions and react automatically to hazards such as smoke or an unsafe temperature, (b) give staff and students a safe, explorable virtual model of the ward for orientation and training, and (c) give facilities or clinical staff a fast, mobile way to look up information about a specific piece of ward equipment simply by pointing a camera at it. A simulation-based project is an appropriate way to demonstrate this concept without the cost, risk, or ethical issues of instrumenting a real ward.

## 1.4 Aim

The aim of this project is to design and demonstrate an integrated Smart Hospital System using IoT, VR, and AR technologies, centered on a two-bed patient ward.

## 1.5 Objectives

- Design an IoT ward network in Cisco Packet Tracer with at least five connected devices and at least two sensors.
- Configure automation conditions that respond to a smoke/fire hazard and to abnormal temperature or humidity.
- Create a complete, navigable VR ward environment in Delightex EDU covering clinical, safety, comfort, and reception areas.
- Program at least five — and ideally many more — labelled, interactive VR objects using CoBlocks so the space can act as a self-guided tour.

- Create a MyWebAR scene containing at least three AR objects with interaction and information access.
- Link the AR scene back to the VR walkthrough so the two immersive layers reinforce each other.
- Explain how IoT, VR, and AR work together as one practical hospital solution, and test the result with documented evidence.

## 1.6 Scope

The project is limited to simulation and prototype visualization. The IoT component uses a Home Gateway, a monitoring laptop, and five registered smart devices inside Cisco Packet Tracer. The VR component represents one complete ward environment containing a two-bed patient room, an en-suite washroom, a reception/entrance area, and three visitor waiting zones. The AR component uses three 3D assets and an on-tap link back to the VR environment to explain selected ward furniture and equipment.

## 1.7 Limitations

- The solution does not use physical sensors or medical devices; real-world electrical, mechanical, or clinical behavior is not measured.
- The VR and AR scenes demonstrate the interaction concept but do not receive live data directly from Packet Tracer.
- Performance depends on the availability of the selected web platforms, browser support, device capability, and internet access.
- The project does not include production-level authentication, encrypted patient-data storage, or a commercial deployment.

## 1.8 Report Structure

Chapter 2 presents the proposed solution and its architecture. Chapters 3, 4, and 5 describe the IoT, VR, and AR simulations in detail, including devices, configuration, and evidence. Chapter 6 explains how the three layers integrate, and documents testing and evaluation. Chapter 7 presents the conclusion, challenges encountered, and future improvements.

# Chapter 2 — Proposed Solution

***System concept, architecture, requirements, and workflow***

## 2.1 System Overview

The proposed solution combines three layers around a single two-bed hospital ward. The IoT layer senses the ward and controls devices automatically. The VR layer provides an explorable, labelled 3D model of the same ward for orientation, demonstration, and training. The AR layer provides a fast, mobile route into equipment information, and — going beyond a simple information card — hands the user directly into the VR tour for a fuller walkthrough. All three layers describe the same physical space from an operational, spatial, and informational point of view respectively.

## 2.2 Proposed System Architecture

| IoT — sensing & control | VR — visualization | AR — mobile access |
|---|---|---|
| Smoke and humidity/temperature sensors | Two-bed ward, washroom, reception, waiting areas | 3D models of reception desk, lamp, laptop |
| Home Gateway evaluates 8 automation conditions | User explores and clicks 26 labelled objects | User views animated models and taps for details |
| Siren, light and fan respond to smoke or heat | Behavior and layout are demonstrated safely | Tap opens the VR ward tour for deeper context |

*Table 2.1 — Integrated Smart Hospital architecture*

## 2.3 Technologies Used

| Technology / Component | Role in the project |
|---|---|
| Cisco Packet Tracer | Networked IoT simulation: device registration, wireless connectivity, monitoring, and automation rules |
| Home Gateway (DLC100) | Central wireless controller and rule engine for the ward's smart devices |
| Delightex EDU | Creation of the navigable 3D ward environment and CoBlocks click-interactions |
| MyWebAR | Creation of the mobile AR scene with 3D models, on-start animation, and tap-to-link actions |
| Laptop / Web browser | Configuration, monitoring, testing, and access to the VR and AR platforms |

## 2.4 Functional Requirements

| ID | Requirement | Evidence / implementation |
|---|---|---|
| FR-01 | Connect at least five IoT devices | Home Gateway, laptop, and five endpoints (smoke detector, humiture monitor, siren, light, fan) shown in the topology |
| FR-02 | Use at least two sensors | Smoke detector plus a combined temperature-and-humidity (Humiture) sensor |
| FR-03 | Apply an automation rule | Eight enabled conditions link the sensors to the siren, light, and fan |
| FR-04 | Provide monitoring and control | Laptop IoT Monitor lists every device and every condition |
| FR-05 | Create one complete VR environment | Ward room, washroom, reception, and three waiting areas were built |
| FR-06 | Provide five interactive/labelled VR objects | Twenty-six clickable objects, each with a titled info panel, are included |
| FR-07 | Create an AR scene with three objects | Reception desk, bedside lamp, and reception laptop models are included |
| FR-08 | Provide AR interaction and information | On-start animation plus a tap action that opens the VR tour are configured |

## 2.5 System Workflow

1. The smoke detector or the humiture (temperature/humidity) sensor observes a change in the ward.
2. The reading is sent wirelessly to the Home Gateway.
3. The Home Gateway evaluates its eight enabled automation conditions against the new reading.
4. The siren, light, and/or fan change state according to the matching rule — a smoke event drives all three together.
5. A member of staff, student, or visitor can separately open the VR ward tour to explore the same space and see how each labelled asset behaves and what it is for.
6. Scanning the AR QR code, or tapping an AR object, opens either the equipment's information card or the VR tour directly, connecting the physical object to its digital counterpart.

## 2.6 Expected Benefits

- Faster response to hazards for patients who cannot act on their own behalf.
- A single smoke event triggers a coordinated response — alarm, lighting, and ventilation together — rather than one isolated action.
- Reduced unnecessary energy use through condition-based fan control.

- Safer orientation and training for new staff or students through a virtual walkthrough instead of the live ward.
- Faster equipment understanding through AR models, animation, and a direct link into the fuller VR tour.
- A reusable simulation that can be extended before any physical implementation.

# Chapter 3 — IoT Simulation Using Cisco Packet Tracer

***Network design, devices, configuration, automation, and monitoring***

## 3.1 IoT Design

Cisco Packet Tracer was used to build the operational layer of the Smart Hospital System. A Home Gateway (DLC100) acts as the wireless access point and rule engine for the ward. A laptop provides access to the IoT Monitor used to register devices and manage automation conditions, while the sensors and actuators are registered as wireless smart-hospital devices around the gateway.

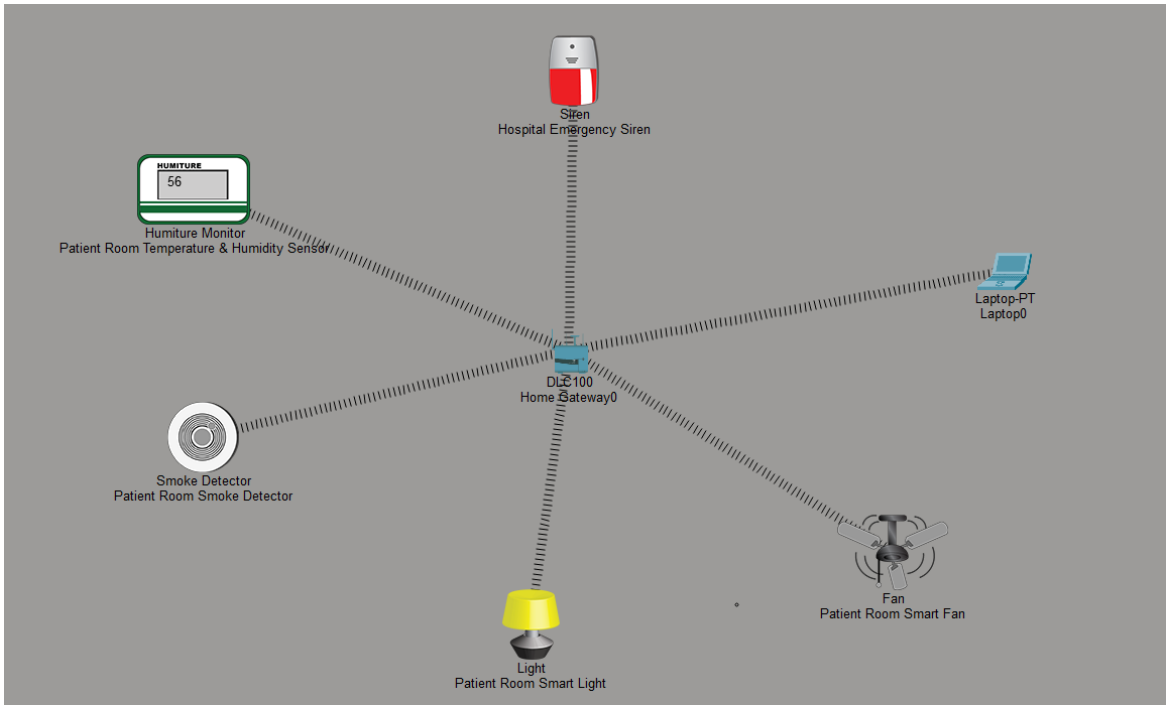



*Figure 3.1: Cisco Packet Tracer topology of the Smart Hospital ward network, showing the Home Gateway wirelessly connected to the monitoring laptop, the smoke detector, the temperature/humidity monitor, the emergency siren, the smart light, and the smart fan.*

## 3.2 Devices Used

| Device | Type | Purpose in the ward |
|---|---|---|
| Home Gateway (DLC100) | Network / controller | Registers every ward device and executes the automation conditions |
| Laptop | Monitoring device | Opens the IoT Monitor and is used to create and edit conditions |
| Smoke Detector | Sensor | Detects a smoke or fire condition in the patient room |
| Humiture Monitor | Sensor | Reads room temperature and humidity for comfort control |
| Hospital Emergency Siren | Actuator | Audibly alerts staff and visitors during a hazard |

| Device | Type | Purpose in the ward |
|---|---|---|
| Smart Light | Actuator | Provides automatic ward lighting during an emergency |
| Smart Fan | Actuator | Provides ventilation to help clear smoke and manage patient comfort |

## 3.3 Network and Device Configuration

7. Place the Home Gateway and the monitoring laptop in the logical workspace.

8. Add the smoke detector, humiture monitor, siren, light, and fan, and rename each device to match its ward function.

9. Configure the Home Gateway's wireless network with WPA2-PSK authentication and AES encryption, and set a shared passphrase.

10. Register each device on the gateway's wireless network and confirm it obtains an IP address.

11. Open the laptop's IoT Monitor and log in to confirm every device is visible.

12. Create and enable the automation conditions described in section 3.4.

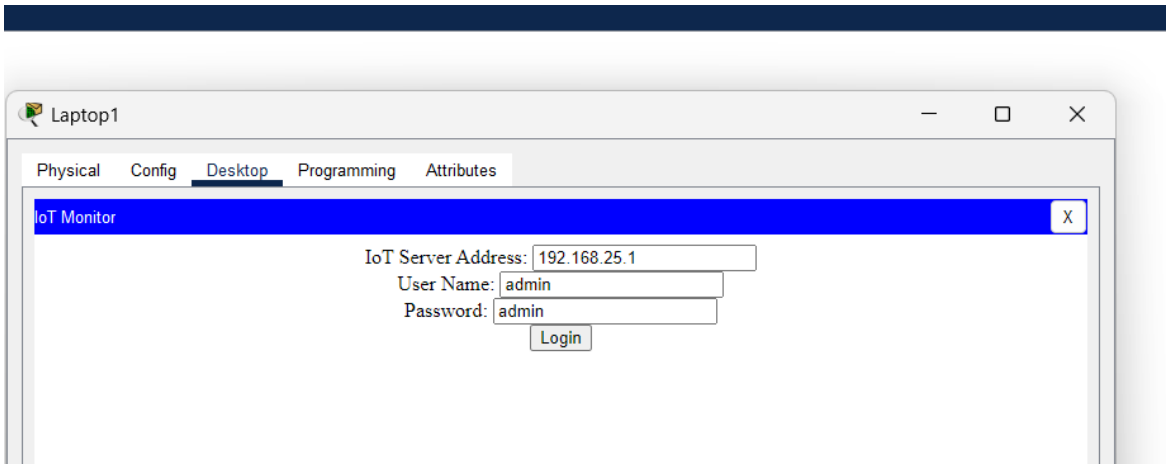



*Figure 3.2: IoT Monitor login screen on the ward laptop, used to sign in to the Home Gateway's device server before viewing devices or conditions.*

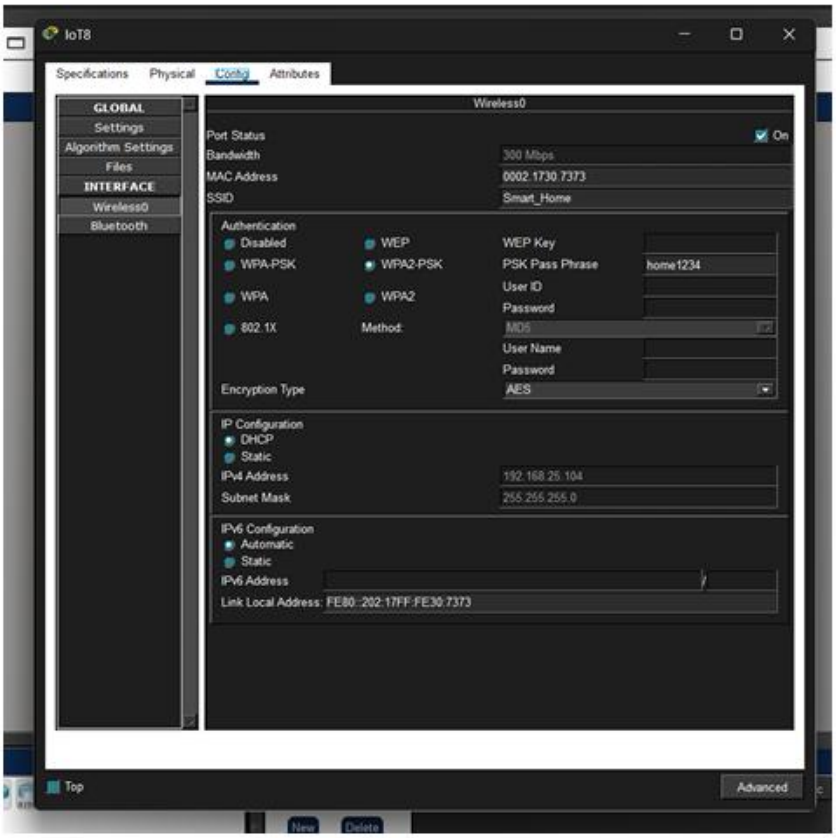



*Figure 3.3: Wireless security configuration on the Home Gateway, showing WPA2-PSK authentication with AES encryption and a DHCsP-assigned IPv4 address for a registered ward device.*

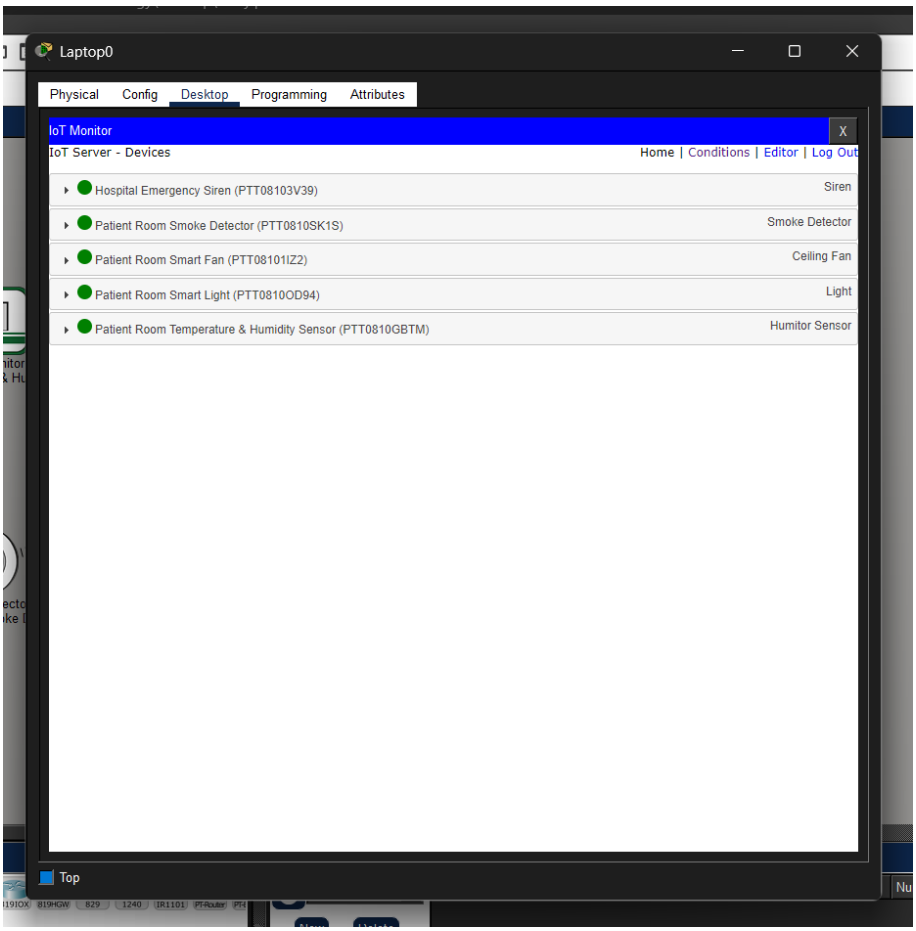



*Figure 3.4: Registered device list on the IoT Monitor, confirming all five ward endpoints — siren, smoke detector, fan, light, and humiture sensor — are online.*

## 3.4 Automation Rules

Eight automation conditions were created and enabled. The smoke-related rules were deliberately split into three paired conditions so that a single smoke event drives three different actuators — siren, light, and fan — together, rather than a single isolated response.

| Rule name | Condition | Action |
|---|---|---|
| Smoke Emergency Alert | Smoke detector reading > 0 | Hospital Emergency Siren switched ON |
| Smoke Normal Siren Off | Smoke detector reading = 0 | Siren switched OFF |
| Smoke Emergency Light | Smoke detector reading > 0 | Smart Light set to full brightness |
| Smoke Normal Light Off | Smoke detector reading = 0 | Smart Light switched OFF |
| Smoke Emergency Fan | Smoke detector reading > 0 | Smart Fan set to High, to help clear the room |
| Smoke Normal Fan Off | Smoke detector reading = 0 | Smart Fan switched OFF |
| High Temperature Fan ON | Humiture reading > 50 | Smart Fan set to High for patient comfort |

| Rule name | Condition | Action |
|---|---|---|
| Normal Temperature Fan OFF | Humiture reading ≤ 50 | Smart Fan switched OFF |

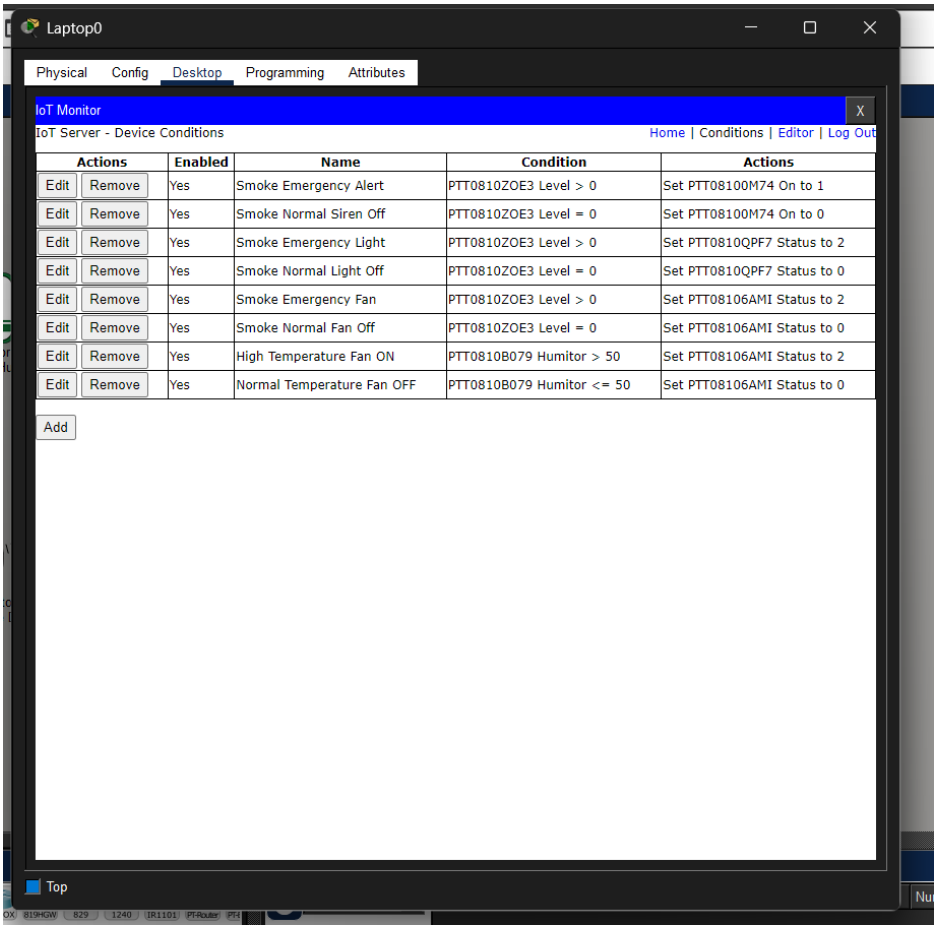



*Figure 3.5: The eight enabled automation conditions in the laptop's IoT Monitor, linking the smoke and humiture sensors to the siren, light, and fan.*

## 3.5 Monitoring and Control

The laptop's IoT Monitor gives a single interface for reviewing every registered device and every automation condition. Each rule has a name, an enabled state, a condition, and an action, so a member of staff could verify how a sensor reading affects the ward's actuators, or adjust the response, without rewiring the network.

## 3.6 Example Operational Scenarios

### *3.6.1 Smoke and Fire Emergency*

When the smoke detector reports a non-zero reading, three rules fire together: the siren sounds, the light switches to full brightness, and the fan is set to High to help disperse smoke. When the smoke condition clears, the matching "Normal" rules return all three devices to their resting state. This coordinated three-device response is more representative of a real fire-alarm system than a single actuator reacting alone.

### *3.6.2 Temperature and Humidity Comfort*

Independently of the smoke rules, the Humiture Monitor drives the fan whenever the combined temperature/humidity reading exceeds 50, and turns it off again once conditions return to normal. This keeps patient comfort automated without interfering with the emergency-response logic.

## 3.7 IoT Outcome

The Packet Tracer simulation exceeds the minimum IoT requirements: five registered smart devices plus the gateway and laptop, two independent sensors, eight enabled automation rules (rather than the minimum of one), a secured wireless network, and a monitoring dashboard. It demonstrates sensing, decision-making, and coordinated multi-device actuation within a single ward network.

# Chapter 4 — Virtual Reality Simulation

***Delightex EDU ward environment, interactive objects, CoBlocks, and navigation***

## 4.1 Environment Design

The VR component was developed in Delightex EDU as a complete hospital ward. The layout contains a two-bed patient room with individual bedside lamps and patient-record monitors, an en-suite washroom with a toilet, sink, and a bathtub with a shower, a reception desk near the main entrance, and three separate visitor/patient waiting zones furnished with sofas, a coffee table, and decorative flower pots. A wall-mounted air-conditioning unit and an oscillating table fan support the ward's climate, and a hospital emergency siren and two fire extinguishers cover safety.

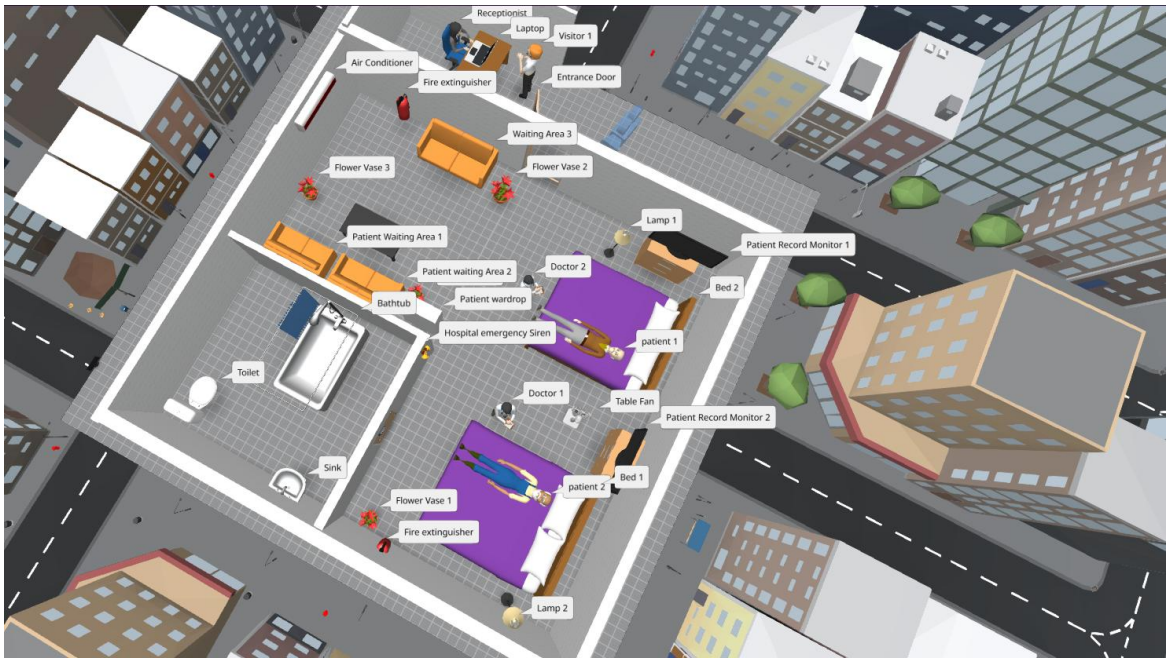



*Figure 4.1: Delightex EDU ward layout in Play mode, with labels identifying beds, patients, doctors, the nurse, monitors, safety equipment, and reception/waiting furniture.*




*Figure 4.2: Delightex EDU editor view of the completed ward, showing the full room layout alongside the asset library used to build it.*

## 4.2 Interactive and Labelled Objects

Every major asset in the ward was made clickable, so the space works as a self-guided orientation tour rather than a static model. The 26 interactive objects are grouped below by function.

| Category | Interactive objects (click → info panel) |
|---|---|
| Clinical care | Bed 1, Bed 2, Patient 1, Patient 2, Patient Record Monitor 1, Patient Record Monitor 2, Doctor 1, Doctor 2, Nurse's patient-storage cabinet |
| Safety & emergency | Hospital Emergency Siren, Fire Extinguisher (reception), Fire Extinguisher (ward) |
| Comfort & climate | Smart Dual-Climate Air Conditioner, Oscillating Table Fan, Patient Bedside Lamp 1, Patient Bedside Lamp 2 |
| Reception & wayfinding | Hospital Main Entrance Door, Reception Desk, Reception Laptop, Visitor 1 |
| Seating & ambience | Three waiting-area sofas, attached washroom (bathtub with shower), four decorative flower vases |

This is well beyond the minimum of five interactive/labelled objects requested in the brief; every clinical, safety, comfort, and reception asset in the ward can be inspected individually.

## 4.3 CoBlocks Programming

CoBlocks were used to give each asset an event-condition-action behavior: "when [object] is clicked → show an information panel with a title and descriptive text." The same When-Play-Clicked structure used for the IoT automation rules in Chapter 3 is mirrored here at the interaction level, so the VR logic and the IoT logic share a consistent event-driven design philosophy.

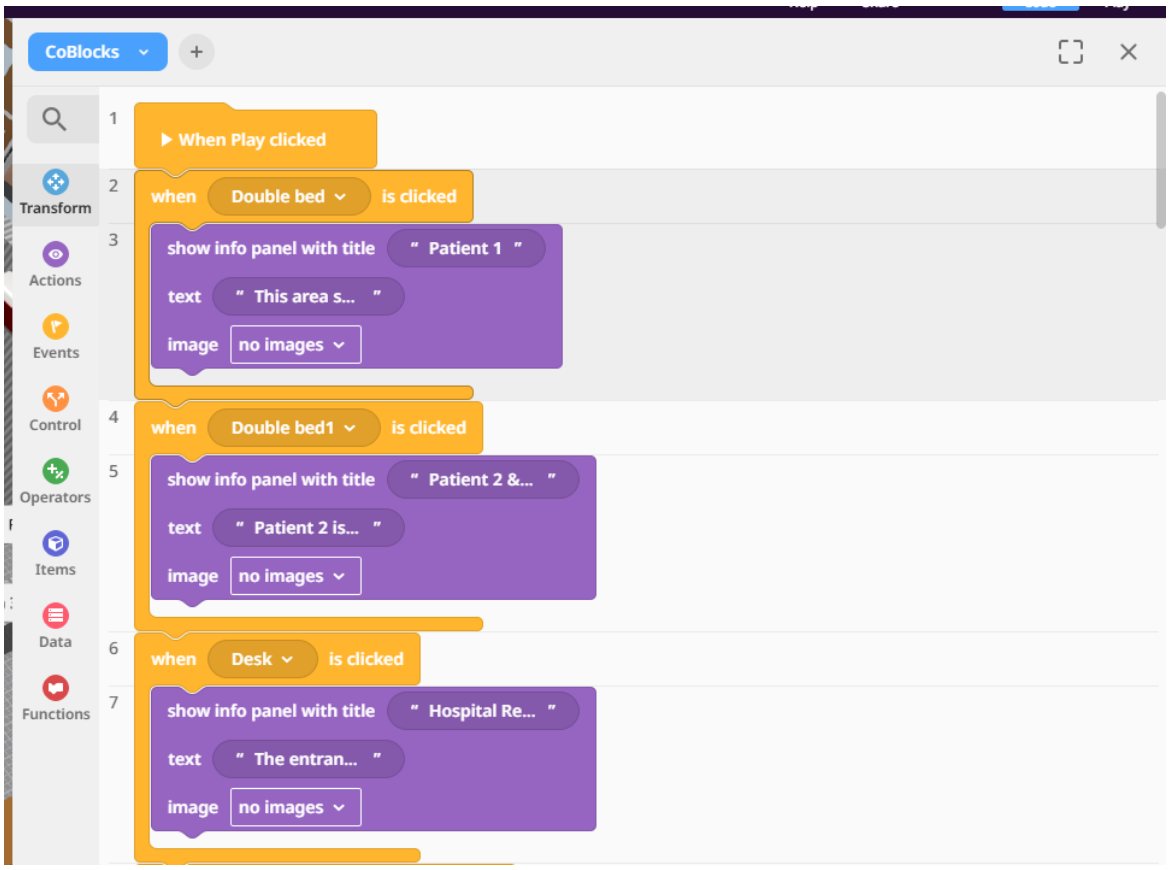



*Figure 4.3: CoBlocks logic for the two patient beds and the reception desk, each opening a titled information panel on click.*

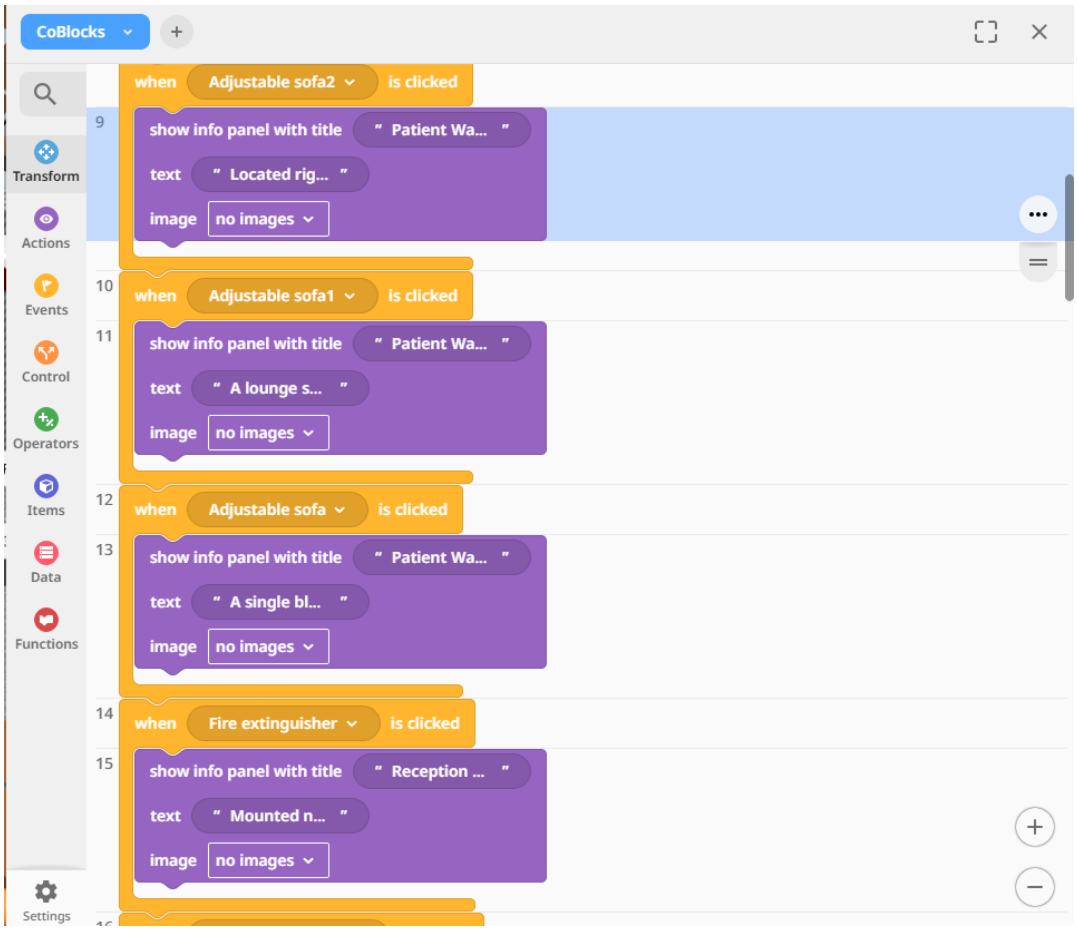



*Figure 4.4: CoBlocks logic for the three waiting-area sofas and the two fire extinguishers.*

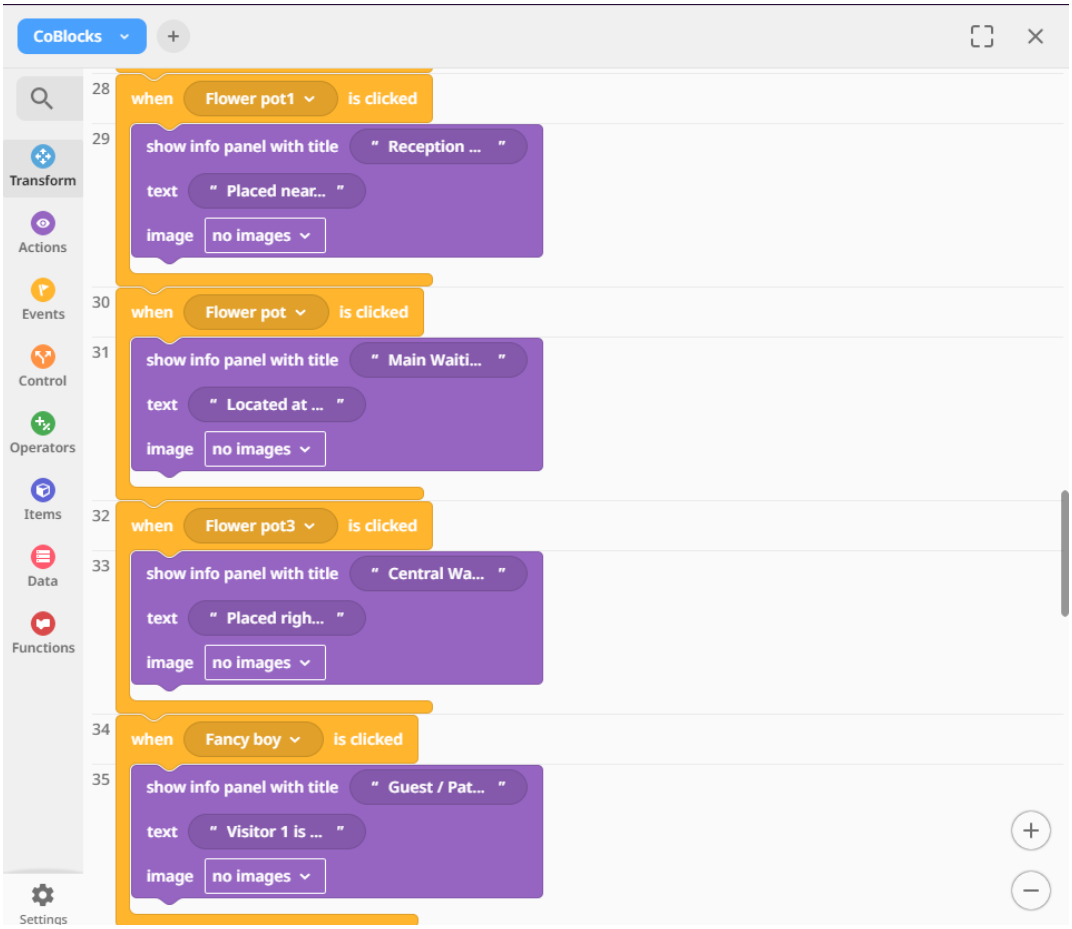



*Figure 4.5: CoBlocks logic for the bathtub-with-shower, the two patient-record monitors (TVs), and the patient storage cabinet.*

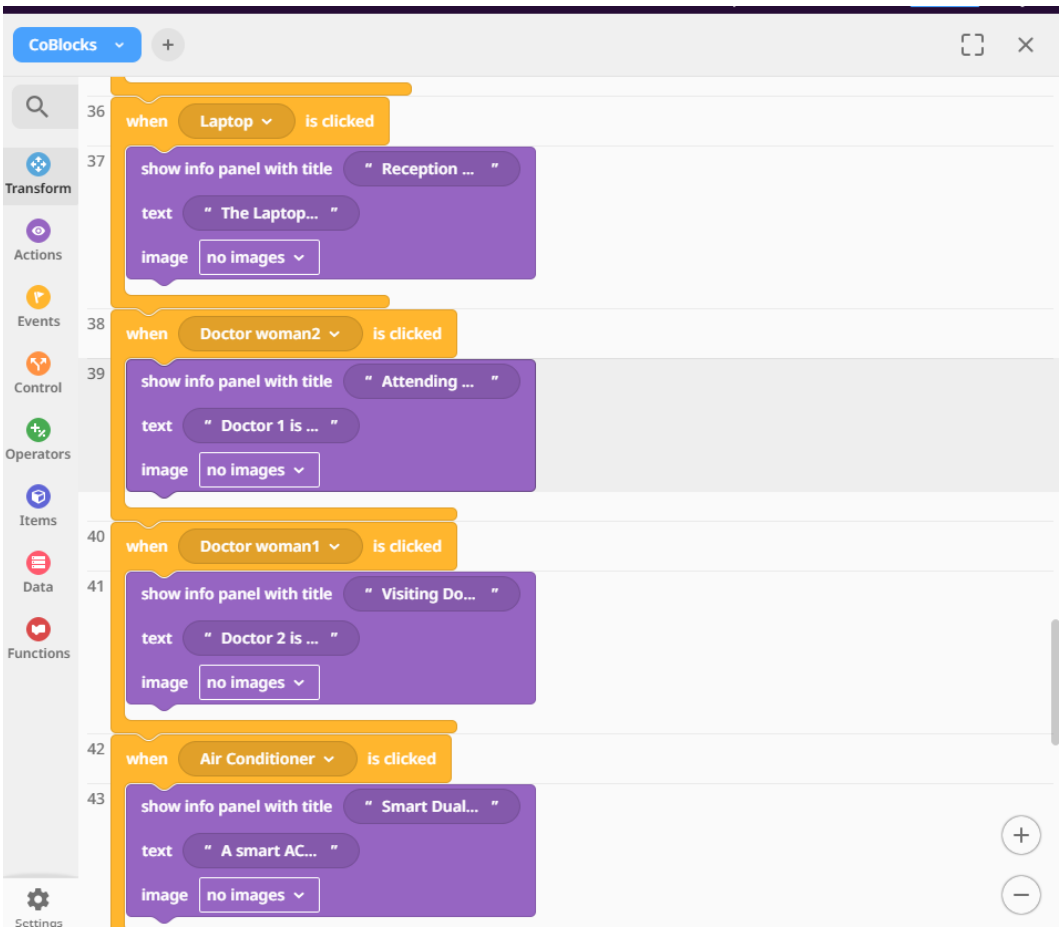



*Figure 4.6: CoBlocks logic for the four decorative flower pots, the visitor character, and the reception laptop.*

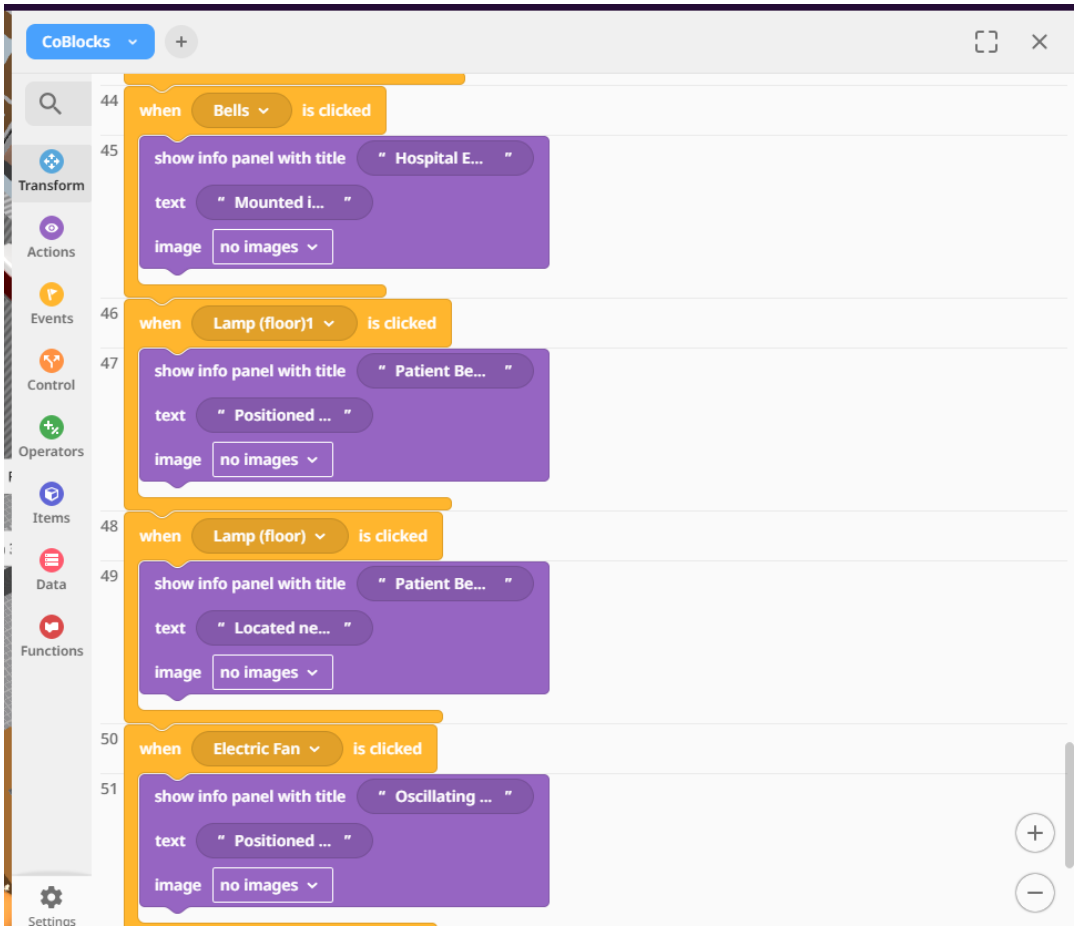


*Figure 4.7: CoBlocks logic for the two doctor characters and the smart air-conditioning unit.*

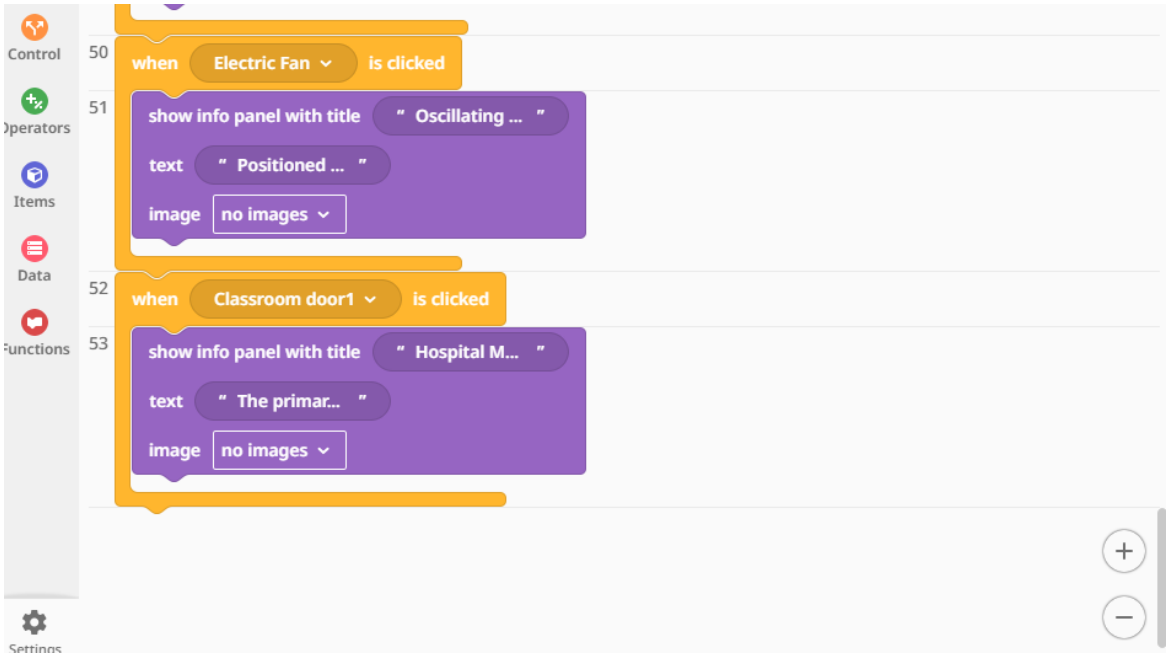



*Figure 4.8: CoBlocks logic for the emergency siren, the two bedside lamps, the oscillating fan, and the main entrance door.*

## 4.4 Navigation and User Experience

In Play mode, the user can walk through the ward and click on any of the 26 labelled assets to open its information panel. A sample of these panels is shown below, illustrating the range of clinical, safety, and comfort information that the walkthrough communicates.

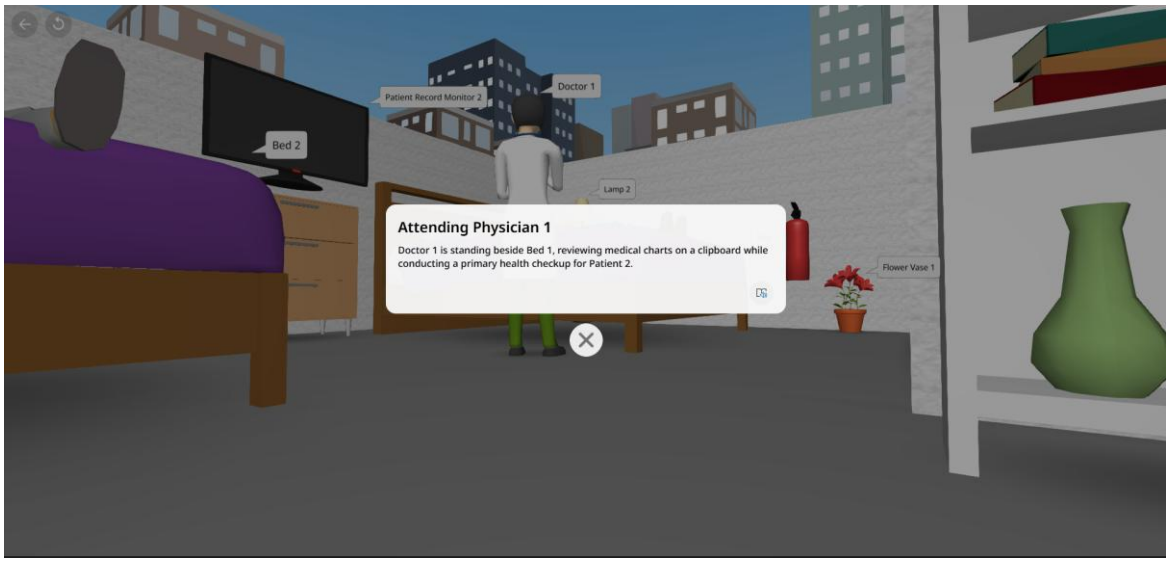



*Figure 4.9: Information panel for the attending doctor, describing the routine checkup being carried out at Bed 1.*

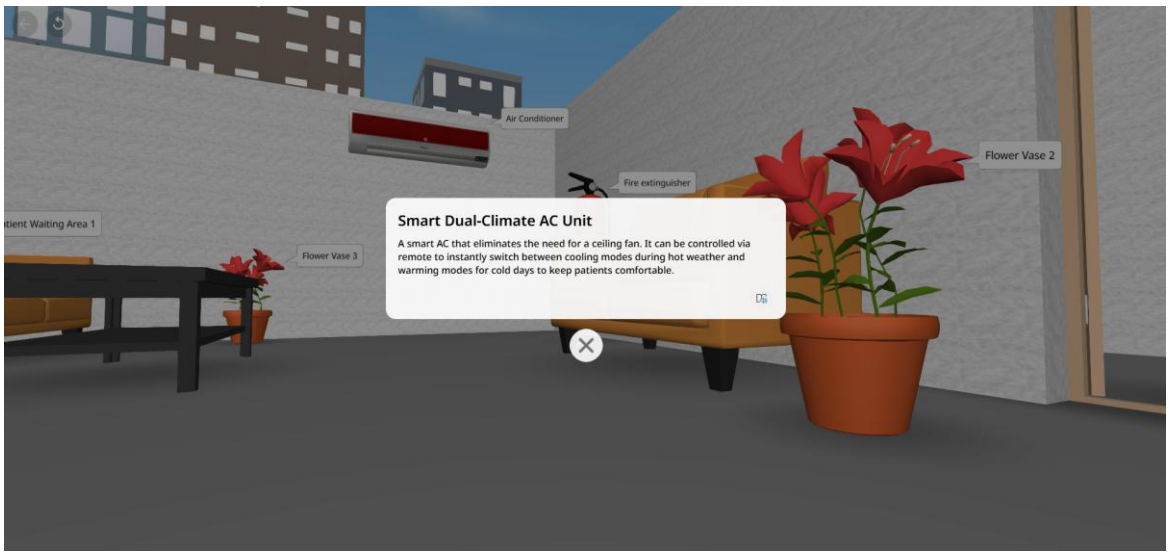



*Figure 4.10: Information panel for the Smart Dual-Climate AC Unit, explaining its remote-controlled cooling and warming modes.*

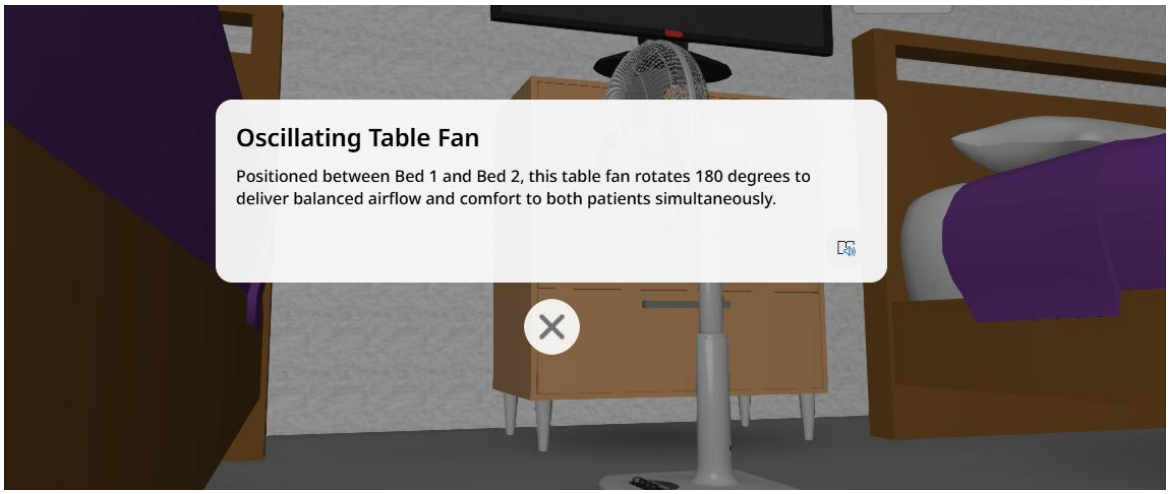



*Figure 4.11: Information panel for the Oscillating Table Fan positioned between the two beds.*

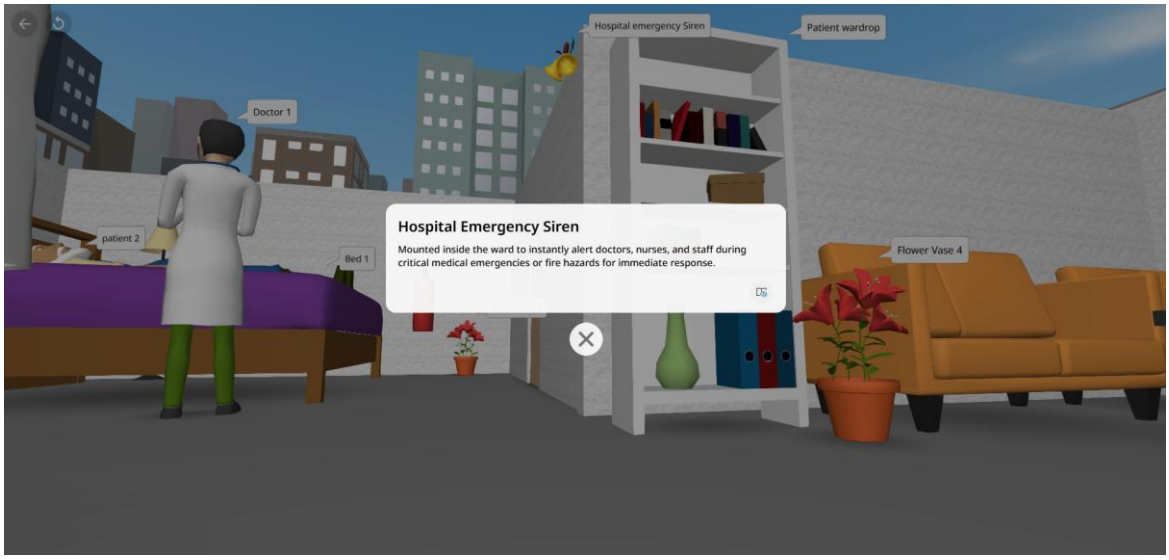



*Figure 4.12: Information panel for the Hospital Emergency Siren, mounted for immediate staff alerting.*

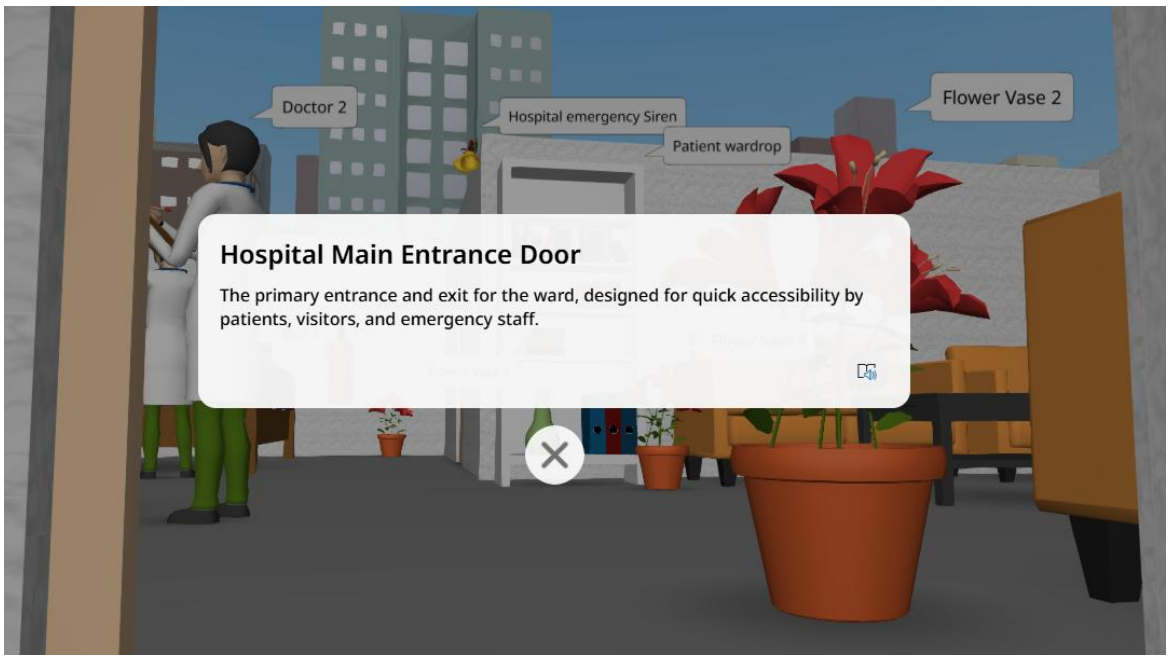



*Figure 4.13: Information panel for the Hospital Main Entrance Door, describing quick accessibility for patients, visitors, and emergency staff.*

## 4.5 VR Outcome

The VR scene exceeds the project's requirements: it is a complete, navigable environment covering clinical, safety, comfort, and reception functions, with 26 labelled and interactive objects — more than five times the minimum requested — each demonstrating a distinct, individually written behavior through CoBlocks. It can be used to explain the ward layout, demonstrate equipment placement, and rehearse emergency responses without any risk to a real patient.

# Chapter 5 — Augmented Reality Simulation

***MyWebAR design, objects, actions, interaction, and QR access***

## 5.1 AR Design

The AR component was created in MyWebAR. The scene includes three 3D assets representing ward furniture and equipment: a Hospital Reception Desk, a Patient Bedside Lamp, and a Reception Laptop. Each object was arranged on a shared 3D workspace and configured with a start-up animation so it visibly comes to life when the AR scene loads.

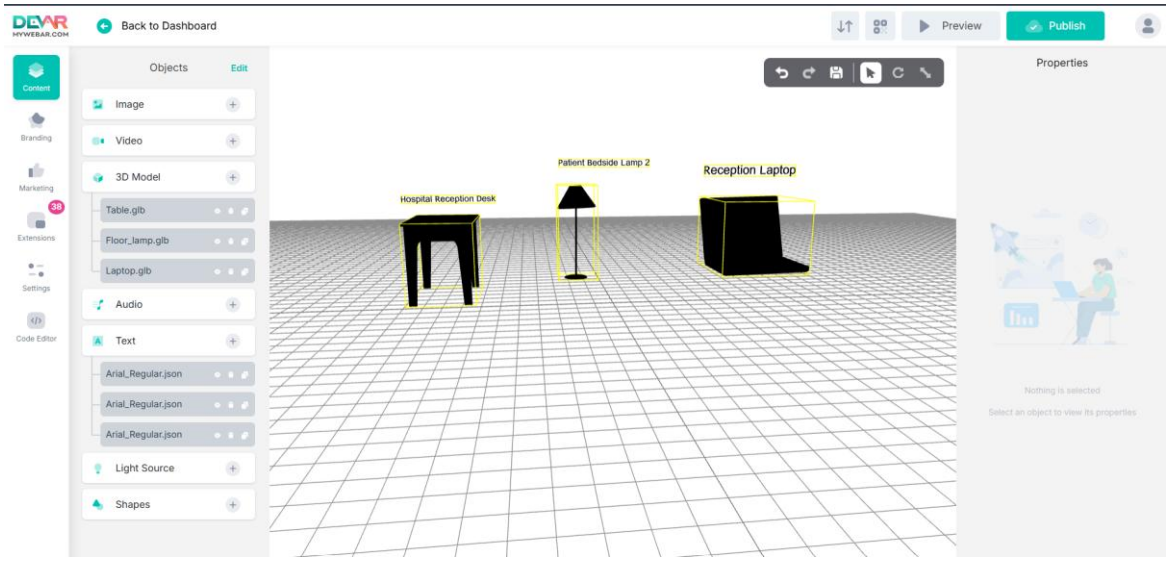



*Figure 5.1: MyWebAR 3D workspace showing the three AR objects — the Hospital Reception Desk, the Patient Bedside Lamp, and the Reception Laptop — positioned and labelled.*

## 5.2 AR Objects and Configured Actions

| AR object | Behavior configured | Purpose |
|---|---|---|
| Hospital Reception Desk | Plays a 3D animation on start | Introduces the reception area of the ward to a mobile viewer |
| Patient Bedside Lamp | Plays a 3D animation on start; tap opens the VR ward tour | Connects a physical lamp to the fuller VR walkthrough |
| Reception Laptop | Plays a 3D animation on start; tap opens the VR ward tour | Let’s a visitor or new staff member jump straight into the ward simulation |

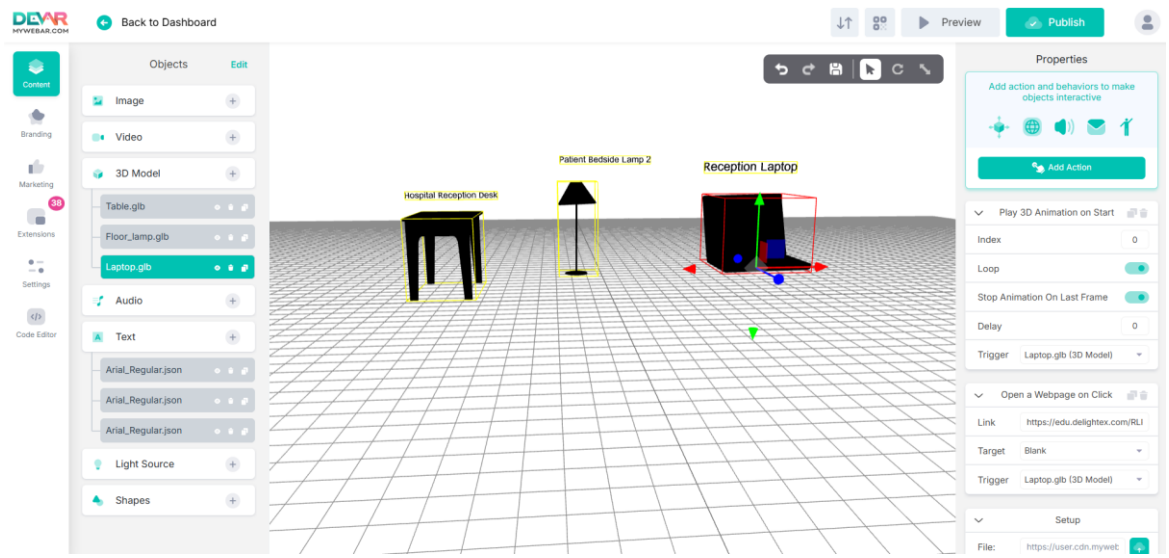



*Figure 5.2: Properties panel for the Reception Laptop object, showing the "Play 3D Animation on Start" and "Open a Webpage on Click" actions, the latter linking to the Delightex EDU ward tour.*

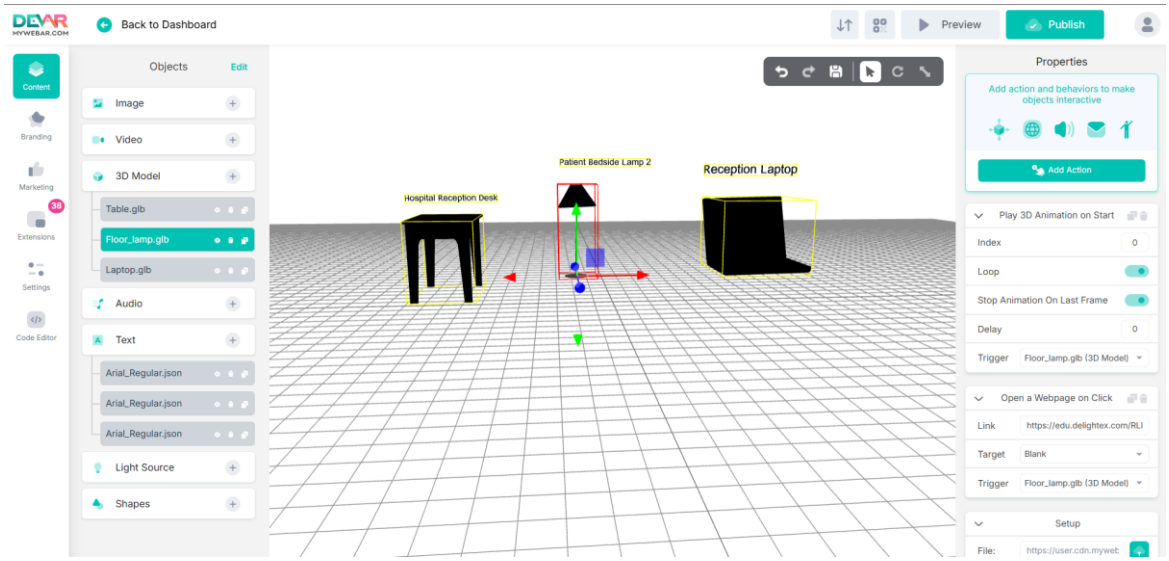



*Figure 5.3: Properties panel for the Patient Bedside Lamp object, configured with the same animation-and-link behavior.*

## 5.3 User Interaction

The user opens the published AR experience by scanning the MyWebAR QR code or following the share link on a phone. Once the scene loads onto the live camera feed, the three models appear anchored to the real surface in front of the user, play their start-up animation, and can be tapped for further action.

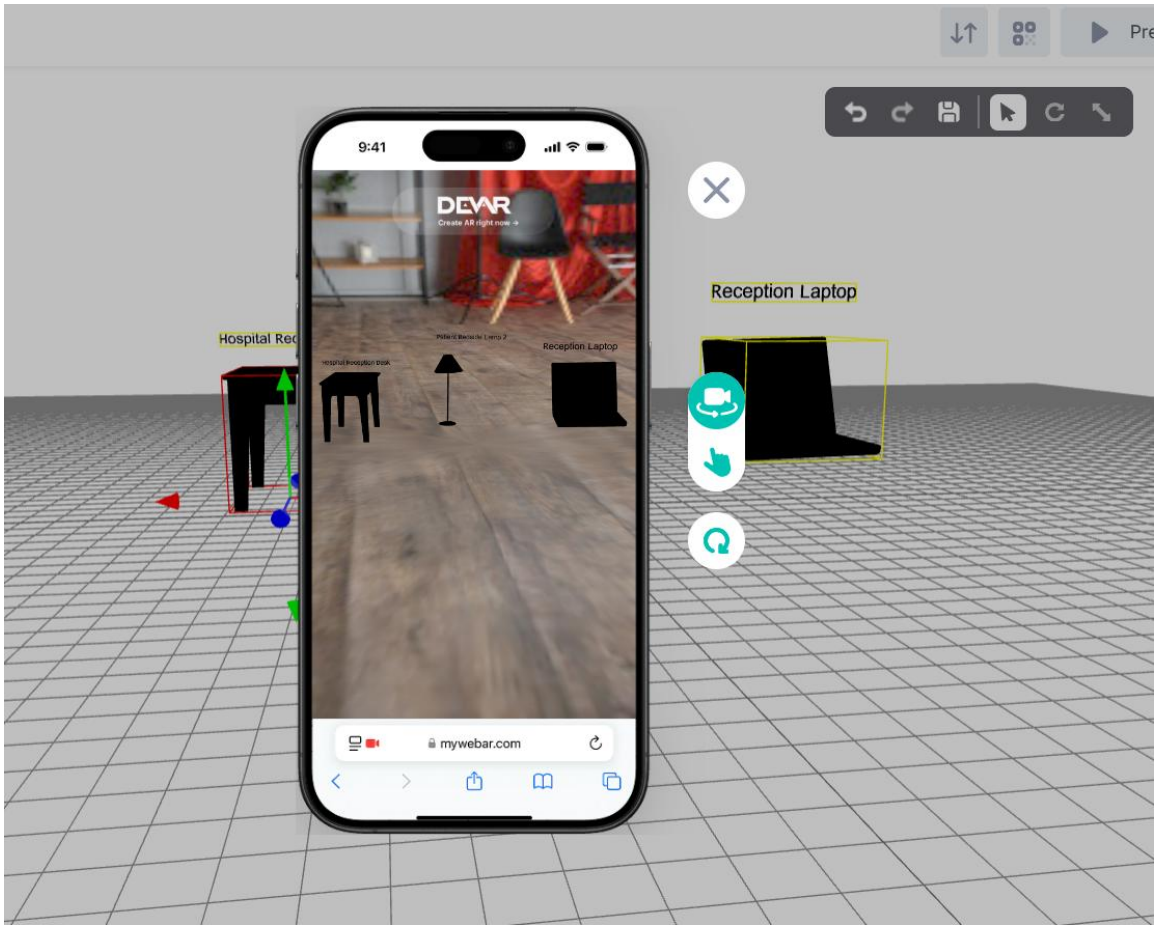



*Figure 5.4: Live camera-anchored AR preview showing the Hospital Reception Desk, Patient Bedside Lamp, and Reception Laptop placed against a real background during on-device testing.*

## 5.4 Linking AR Back into the VR Tour

Rather than only showing a static text overlay, the Reception Laptop and Patient Bedside Lamp were both configured with an "Open a Webpage on Click" action pointed at the Delightex EDU share link. This means a visitor who taps either object in the real world is taken directly into the full VR ward walkthrough from Chapter 4, rather than a short caption. This is the concrete integration point between the AR and VR layers described in Chapter 6.

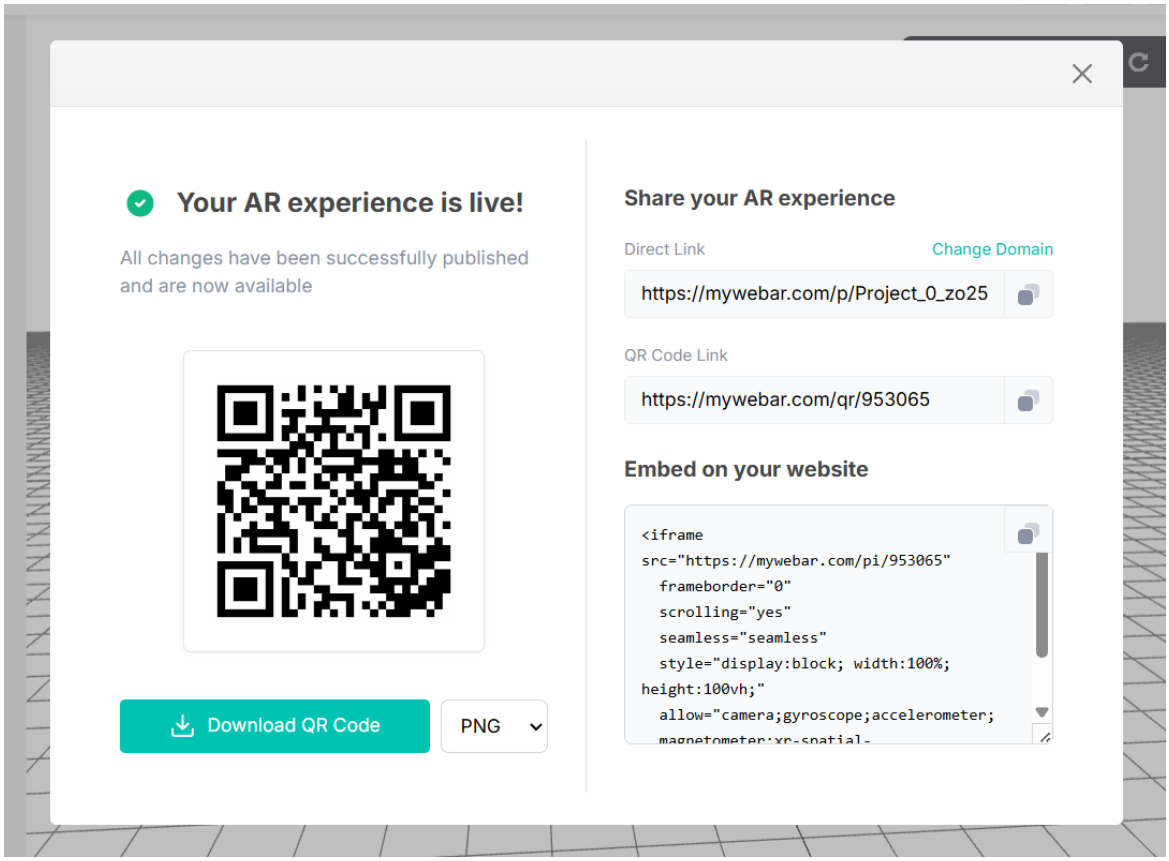



*Figure 5.5: Published AR experience showing the share link and QR code generated after publishing the MyWebAR scene.*

## 5.5 AR Outcome

The AR scene meets and extends the minimum project requirements: it provides one scene with three 3D objects (rather than the minimum three treated as separate static cards), animation on load, and — beyond a simple information overlay — a direct tap-through into the VR experience, plus QR-based access for quick mobile testing.

# Chapter 6 — Integration, Testing, and Evaluation

*How the three components work together and how the prototype was verified*

## 6.1 Integration of IoT, VR, and AR

| Technology | Primary contribution | Connection to the complete solution |
|---|---|---|
| IoT | Sensing, monitoring, rules, and coordinated actuation | Defines what the ward does automatically during a real hazard or comfort change |
| VR | Immersive, 26-object environment for navigation and training | Shows where every device is located and demonstrates its role without touching real equipment |
| AR | Mobile 3D models, animation, and a direct link into VR | Bridges a physical object in front of the user to the fuller VR walkthrough |

## 6.2 Integrated User Journey

13. A ward condition — smoke, or an abnormal temperature/humidity reading — is detected by a sensor.

14. The Home Gateway evaluates its automation conditions and changes the siren, light, and/or fan accordingly.

15. The same equipment and its behavior are represented in the VR ward for safe, self-guided exploration and staff training.

16. A visitor or new staff member scans the AR QR code and sees the reception desk, lamp, and laptop rendered live on their own camera feed.

17. Tapping the AR lamp or laptop opens the full VR ward tour in the browser, carrying the user from a quick mobile glance into the deeper 26-object walkthrough.

18. The combined experience improves hazard response time, staff orientation, and visitor/patient understanding of the ward.

## 6.3 Test Plan and Results

| ID | Test objective | Expected result | Status |
|---|---|---|---|
| T01 | Verify IoT device connectivity | All five devices connect wirelessly to the Home Gateway | Pass |
| T02 | Verify smoke emergency automation | Siren, light, and fan all respond together to a smoke reading | Pass |
| T03 | Verify humidity/temperature automation | Fan switches on above the configured threshold and off below it | Pass |
| T04 | Verify wireless security configuration | Home Gateway wireless tab shows WPA2-PSK with AES encryption | Pass |

| ID | Test objective | Expected result | Status |
|---|---|---|---|
| T05 | Verify VR environment loads | Complete ward interior loads and is navigable in Play mode | Pass |
| T06 | Verify VR interactive objects | 26 objects each open a distinct, titled information panel | Pass |
| T07 | Verify CoBlocks logic | Click events are wired for clinical, safety, comfort, and reception assets | Pass |
| T08 | Verify AR scene objects | Reception desk, lamp, and laptop load with a start animation | Pass |
| T09 | Verify AR-to-VR link | Tapping the AR lamp/laptop opens the Delightex ward tour | Pass |
| T10 | Verify QR / share access | Delightex share link and MyWebAR QR/link both resolve correctly | Pass |

## 6.4 Requirement Compliance

The evidence confirms the prototype exceeds the minimum IoT device and sensor requirement, includes eight automation conditions against a minimum of one, and offers full monitoring through the laptop's IoT Monitor. The VR environment is complete, navigable, and contains 26 labelled/interactive objects against a minimum of five. The AR scene contains three 3D objects with animation, tap interaction, a direct link back into the VR tour, and QR-based access.

## 6.5 Benefits of the Integrated Solution

- Patient safety: a single smoke event triggers a coordinated siren, lighting, and ventilation response, which matters most for patients who cannot self-evacuate.
- Comfort: humidity/temperature-based fan control keeps the ward comfortable without manual intervention.
- Energy efficiency: the fan and light remain off when their triggering condition is absent.
- Safer training: VR lets staff or students learn ward layout and equipment behavior without touching real patients or equipment.
- Faster orientation: AR gives visitors and new staff an immediate, camera-based introduction to key ward furniture, with a direct route into the deeper VR tour.
- Design validation: building the simulation surfaced placement and workflow issues (see Chapter 7) before any physical installation would begin.

## 6.6 Risk and Ethical Considerations

- Patient Record Monitor data is sensitive; any real deployment would need informed consent and access controls before displaying patient information.
- Authentication and encrypted communication — beyond the WPA2-PSK link used here — would be required before connecting a real ward network to any cloud service.
- Emergency automation must include a manual override and fail-safe behavior so staff can act if a rule misfires.
- A sensor fault or network outage should never silently disable the smoke or humidity rules without an alternative alert.

- VR and AR information must stay accurate and up to date so that staff training does not rely on misleading instructions.

# Chapter 7 — Conclusion and Future Work

***Project outcomes, challenges, lessons, and extensions***

## 7.1 Conclusion

This project demonstrates an integrated Smart Hospital System using IoT, VR, and AR, built around the specific needs of a patient ward rather than a general-purpose space. Cisco Packet Tracer provides the operational simulation, with a wirelessly secured Home Gateway coordinating five devices through eight automation conditions that respond to both smoke and temperature/humidity hazards. Delightex EDU provides a richly interactive virtual ward — 26 individually programmed objects — that turns a walkthrough into a self-guided orientation tour. MyWebAR provides a mobile, camera-anchored entry point that links straight back into that same VR tour rather than stopping at a static caption.

Together, the three components show how emerging technologies can improve patient safety, staff training, and visitor understanding in a hospital setting, and that simulation tools can be used to design, explain, and test a smart-ward concept before any physical implementation.

## 7.2 Challenges Encountered

- Renaming and correctly matching generic Packet Tracer devices to hospital-specific roles (siren, humiture sensor, smart fan) while keeping every automation condition pointed at the right device.
- Designing a smoke response that drives three actuators together, rather than one rule per device, without creating conflicting states.
- Arranging 26 closely spaced VR objects and their labels inside a two-room ward without visual clutter.
- Writing and testing 26 separate CoBlocks click events without duplicate or conflicting object names.
- Configuring the MyWebAR "Open a Webpage on Click" action so it reliably opens the correct Delightex share link on a mobile device.
- Keeping the Delightex and MyWebAR share links stable across several rounds of edits and republishing.

## 7.3 Lessons Learned

- A hospital automation rule is safer when a single hazard condition is deliberately fanned out to every relevant actuator, rather than left as one isolated action.
- Interactivity scales the training value of a VR environment — moving from five labelled objects to 26 turned a demonstration into a usable self-guided tour.
- AR is most useful for a ward not as a stand-alone information card, but as a fast bridge into a fuller VR resource.
- Screenshots, structured tables, and clearly named automation rules are essential evidence for demonstrating that a simulation actually meets its stated requirements.

## 7.4 Future Improvements

- Connect a physical microcontroller and real sensors to validate the automation logic in an actual room.
- Add nurse-call mobile alerts triggered by the same smoke and humidity conditions.
- Use RFID or QR-based patient identification inside the AR scene to pull up the correct Patient Record Monitor.
- Add a live bed-occupancy dashboard fed from the VR/IoT layers for ward management.
- Apply secure authentication, encrypted communication, role-based access, and audit logging before handling real patient data.
- Scale the design from one ward to a multi-ward floor plan and evaluate performance with real nursing staff.

## 7.5 Final Summary

The completed prototype is a strong foundation for a more advanced smart-hospital platform. Its main contribution is not any single simulation on its own, but a coordinated demonstration of automatic sensing and response through IoT, safe immersive training through VR, and a fast, linked mobile bridge between the two through AR.

# References

[1] Cisco Networking Academy. (n.d.). Cisco Packet Tracer resources. https://www.netacad.com/cisco-packet-tracer

[2] Delightex GmbH. (n.d.). Delightex EDU. https://edu.delightex.com/

[3] MyWebAR. (n.d.). Web augmented reality platform. https://mywebar.com/

[4] International Campus of Science and Technology. (2026). Emerging Technologies — Integrated Simulation Project guidelines.

[5] International Campus of Science and Technology. (2026). Instructions for final submission — Emerging Technologies.

# Appendix A — Submission Evidence

## A.1 Included Evidence

| Item | File / evidence |
|---|---|
| Project report | Smart_Hospital_Emerging_Technologies_Report.docx |
| Cisco Packet Tracer source | [add your saved .pkt file name here] |
| IoT evidence | Topology, wireless configuration, device list, and automation screenshots in Chapter 3 |
| VR evidence | Environment, CoBlocks, and information-panel screenshots in Chapter 4 |
| AR evidence | Workspace, action configuration, live-preview, and QR screenshots in Chapter 5 |

## A.2 Final Submission Checklist

- Rename the submission folder using the required format: Index Number_Name.
- Place the project report PDF/DOCX, the Cisco Packet Tracer file, and this links document inside the folder.
- Upload the demonstration video to OneDrive and set access to "Anyone with the link can view".
- Confirm the Delightex EDU and MyWebAR links below open correctly before submitting.
- Compress the folder as a ZIP file and upload it to OneDrive.
- Prepare the final form document with your index number, name, lecturer name, and ZIP link.
- Submit the hard copy and online materials within the announced deadline.

## A.3 Project Links

| Required link | Entry |
|---|---|
| OneDrive demonstration video link | Recording 2026-07-30 032449.mp4 |
| Delightex EDU (VR) project link | https://edu.delightex.com/RLP-BQU |
| MyWebAR (AR) project link | https://mywebar.com/p/Project_0_zo25pog188 |

*Note: confirm both links are set to the correct sharing permissions and open correctly before final submission.*